PRIYANKA GAWADE



Machine Learning Researcher | Astrophysicist | Statistical Inference | Physical Modelling

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PROFESSIONAL PROFILE

Machine learning researcher with a PhD in astrophysics and a B.Tech in Mechanical Engineering, currently a postdoctoral researcher at the University of Rome Tor Vergata. Research experience spans physical modelling, statistical inference, uncertainty quantification, time series analysis, astronomical imaging, and validation on simulated and real data, supported by peer-reviewed publications and international conference presentations. Designed and implemented Convolutional Neural Network (CNN) and Bayesian Neural Network (BNN) architectures for galaxy-scale strong lensing inference and explored denoising using autoencoders. Proficient in Python, TensorFlow, and Keras; currently developing machine learning methods for transient detection and classification from light curves and images in Rubin LSST data.

CORE EXPERTISE RELEVANT TO SINTEF ANALYTICS & AI

Agentic AI and agentic programming	Practical use of Codex to support development, testing, debugging, and iteration of scientific data analysis pipelines
Modelling physical systems	Physical and mathematical modelling of galaxy-scale strong lensing systems; machine learning parameter inference tested on simulated and real astronomical data
Statistical inference and uncertainty quantification of AI models	Bayesian neural networks; probabilistic parameter inference; uncertainty and covariance estimation
Machine learning on multi-modal data	Transient light curves (time series data) and astronomical images for detection and classification; experience relevant to multi-modal machine learning
Programming & Frameworks	Python, TensorFlow, Keras, Linux, LaTeX
Research Track Record	Peer-reviewed publications; international conference presentations; scientific writing, collaboration, and communication

RESEARCH & MACHINE LEARNING EXPERIENCE

Postdoctoral Researcher — University of Rome Tor Vergata, Rome, Italy **2025–Present**

- Developing machine learning algorithms for transient detection and classification in large Rubin LSST astronomical datasets using transient light curves (time series data) and astronomical images.
- Using agentic AI coding tools such as Codex to support development, testing, debugging, and iteration of scientific data analysis pipelines.
- Contributing to collaborative research in Rubin LSST and high-energy transient science.

PhD Researcher — IUCAA, Pune, India **2020–2025**

- Designed and implemented CNN architectures to infer physical parameters of galaxy-scale strong gravitational lenses from Hyper Suprime-Cam imaging data.
- Developed BNN models to estimate lens and source parameters together with their uncertainties and covariances.
- Used multi-channel astronomical imaging for physical parameter inference.
- Evaluated machine learning models on both simulated and real astronomical data and investigated discrepancies between simulations and real observations.
- Explored denoising methods using autoencoders for astronomical imaging data.
- Investigated gravitational lensing probes of dark matter, including primordial black hole constraints using GRB microlensing parallax.

EDUCATION

PhD, Astrophysics — IUCAA, Pune, India **2020–2025**
Graduate School — IUCAA, Pune, India **2019–2020**
Integrated MSc-PhD — IISER Pune, Pune, India **2017–2019**
B.Tech, Mechanical Engineering — VIT Pune, Pune, India **2012–2016**

SELECTED PEER-REVIEWED PUBLICATIONS

- **Gawade et al. (2026):** Bayesian neural network prediction of lens and source parameters along with their uncertainties for strong lensing samples from Hyper Suprime-Cam Survey. Under review in MNRAS.
- **Gawade et al. (2025):** Neural network prediction of model parameters for strong lensing samples from Hyper Suprime-Cam Survey. MNRAS. DOI: 10.1093/mnras/staf935.
- **Gawade et al. (2023):** On the feasibility of primordial black hole abundance constraints using lensing parallax of GRBs. MNRAS. DOI: 10.1093/mnras/stad3336.
- **Bhalerao, Sawant, ..., Gawade et al. (2024):** Science with the Daksha High Energy Transients Mission. Experimental Astronomy 57, 23. DOI: 10.1007/s10686-024-09923-1.

SELECTED PRESENTATIONS & RESEARCH COLLABORATIONS

Rubin LSST | Daksha Science

Sep 2025 Bayesian neural network prediction of lens and source parameters along with their uncertainties for strong lenses from HSC — LSST@Europe7.

Jan 2025 Bayesian neural network prediction of model parameters and uncertainties for strong lenses from HSC — AI/ML in Astronomy and Astrophysics Workshop, IUCAA, India.

Mar 2024 Neural network prediction of model parameters of strong lenses from HSC — Rubin LSST Strong Lensing Meeting, University of Oxford, UK.

Mar 2024 PBH dark matter constraints using microlensing parallax of gamma-ray bursts — Multi-messenger Gravitational Lensing, Royal Society Meeting, Manchester, UK.

TEACHING, SCIENCE OUTREACH & LANGUAGES

- Assisted with tutorials and grading for an MSc Cosmology course taught by Prof. Surhud More, University of Pune (Spring 2022).
- Contributed to science popularization through public talks, posters, and gravitational lensing demonstrations at IUCAA.
- Strong English proficiency; experienced in international teamwork, scientific presentations, and communicating science to specialist and general audiences.

Languages: English | Hindi | Marathi